

# Topographic Radar Geometry-Guided Cross-Attention for Alpine Avalanche Debris Mapping in Sentinel-1 SAR: Cross-System Generalization Across the Alps, Pamirs, and Arctic

Akshay Sharma<sup>1,\*</sup>, Lalji Prasad<sup>1</sup>

<sup>1</sup>Department of Computer Science and Engineering, SAGE University, Indore 452020, Madhya Pradesh, India

\*Corresponding Author: Akshay Sharma ([akssha74@gmail.com](mailto:akssha74@gmail.com))

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- Akshay Sharma: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.
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The AvalCD benchmark dataset is publicly accessible on Zenodo (DOI: [10.5281/zenodo.15863589](https://doi.org/10.5281/zenodo.15863589)). All code and replication suites are public at <https://github.com/akssha74/TopoRadar-Net-AvalCD>. Publication-bound checkpoints, the formatted anonymous Word manuscript, and all artifacts are distributed in versioned post-freeze Release v1.0.15: <https://github.com/akssha74/TopoRadar-Net-AvalCD/releases/tag/v1.0.15>.